

Toward a Quantitative $c_{\text{eff}}(a)$: Parameterization, Constraints, and Degeneracy-Breaking Tests

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1. Motivation and Status

The four-channel consistency test (DOI: 10.6084/m9.figshare.31304980) established that EFC's grid coupling enters cosmology through the background expansion sector, not through the Poisson equation. The key evidence:

- **Poisson channel excluded:** $\mu > 1$ produces growth *enhancement* ($\chi^2 = 18.2$ on BOSS RSD), wrong direction for S_8 tension.
- **Background channel favored:** $\beta \cdot T(a)$ in $H^2(a)$ produces growth *suppression* via Hubble friction ($\chi^2 = 1.09$), correct direction.
- **Same $\beta = 0.08$** simultaneously improves BAO distances and growth rates (combined $\Delta\chi^2 = -3.89$).

If the grid modifies the vacuum sector that governs $H(a)$, then the effective properties of spacetime-as-medium change with cosmic state. In any medium, the propagation speed depends on the medium's properties. Therefore, a weak state-dependence of the cosmological propagation limit (c_{eff}) follows as a consistency requirement, not as an independent hypothesis.

Current epistemic status:

- ☐ Conceptually motivated (background channel identification)
- ☐ Internally consistent (no contradiction with existing tests)
- ☐ Indirectly supported (four channels favor the mechanism)
- ☐ No unique smoking-gun signature identified
- ☐ No direct detection of $\delta c/c$
- ☐ Not yet distinguished from $w_0 w_a \text{CDM}$

2. The Parameterization

2.1 Ansatz

We write:

$$c_{\text{eff}}(a) = c_0 [1 + \varepsilon \cdot \beta \cdot T(a)]$$

This is a first-order Taylor expansion in the grid response; higher-order terms $O((\beta T)^2)$ are neglected. The expansion is valid for $\varepsilon \cdot \beta \cdot T \ll 1$, which is satisfied for the realistic parameter range $\varepsilon \lesssim 0.1$. We treat $\varepsilon \beta T$ as a small, dimensionless control parameter analogous to a linear response coefficient in an effective medium description.

The parameters are:

- c_0 — the laboratory speed of light (unchanged locally)
- $\beta = 0.08$ — the background coupling amplitude from BAO+RSD
- $T(a)$ — the standard EFC transition function, normalized to peak at unity ($z_{\text{peak}} = 0.441$), with $T \rightarrow 0$ when $\Omega_m \rightarrow 1$

- ϵ — a new dimensionless coefficient setting the strength of the propagation-limit coupling relative to the expansion coupling

2.2 Physical meaning of ϵ

The parameter ϵ encodes how strongly the vacuum-sector modification that changes $H(a)$ also changes the effective propagation speed. Three limiting cases:

ϵ value	Physical meaning
$\epsilon = 0$	No c_{eff} variation; grid modifies expansion but not propagation (pure dark energy equivalent)
$\epsilon = 1$	Full coupling: same fractional change in c_{eff} as in the Λ -sector contribution to H^2
$\epsilon \ll 1$	Weak coupling: propagation limit responds subdominantly to the vacuum state change

EFC's internal logic suggests $\epsilon \sim \mathcal{O}(1)$, because the same underlying grid state governs both the vacuum energy density ($\rightarrow H(a)$) and the vacuum's effective elastic/causal properties ($\rightarrow c_{\text{eff}}$). However, this is not proven; ϵ could be suppressed by the specific coupling structure.

2.3 Numerical values at $\beta = 0.08$

Default realistic range: $\epsilon \sim 0.01\text{--}0.1$

Redshift z	$T(z)$	$\beta \cdot T$	$\delta c/c$ ($\epsilon = 0.1$)	$\delta c/c$ ($\epsilon = 0.01$)
0.0	0.87	0.070	0.70%	0.070%
0.44 (peak)	1.00	0.080	0.80%	0.080%
0.51 (BOSS)	0.99	0.079	0.79%	0.079%
1.0	0.69	0.055	0.55%	0.055%
2.0	0.22	0.018	0.18%	0.018%
3.0 ($\text{Ly}\alpha$)	0.107	0.0086	0.086%	0.0086%
1100 (CMB)	~ 0	~ 0	~ 0	~ 0

Why $\epsilon \gg 1$ is excluded: If c_{eff} enters the distance integral as $d_C = \int c_{\text{eff}}/H \, dz$, then $\epsilon = 1$ produces $\sim 5\text{--}8\%$ coherent distance shifts over $z \sim 0\text{--}1$. This would be harshly penalized by combined SN+BAO fits. The four-channel consistency at $\beta = 0.08$ already implicitly constrains the total distance effect, bounding ϵ from above.

3. Observational Consequences

3.1 Modified distance relations

If c varies, the comoving distance becomes:

$$d_C(z) = \int_0^z c_{\text{eff}}(z') / H(z') \, dz'$$

Here c_{eff} is treated as a physical propagation speed relative to the cosmological frame defined by comoving observers, not a coordinate redefinition. Varying c_{eff} represents a physical change in signal propagation, not a gauge artifact.

This modifies both D_L and D_A by the same factor, preserving the cosmic distance duality relation (CDDR): $D_L = (1+z)^2 D_A$. **This applies only to the metric implementation (Case i); in a medium interpretation (Case ii), the relation can break.**

Critical point: In the metric implementation, c_{eff} variation is *degenerate* with a modified $H(z)$ in standard distance tests. This is the core degeneracy problem for Case (i).

3.2 Two implementations: metric vs optical-medium

Critical distinction. The physical meaning of $c_{\text{eff}}(a)$ determines which observational signatures follow. EFC has not yet committed to one implementation; both must be analyzed.

Case (i): Metric VSL (geometric implementation)

c_{eff} enters as a modified conformal structure: photons follow null geodesics of an effective metric $g_{\text{eff}}(S)$, where the grid state S determines the local lightcone opening. In this case:

- **CDDR is preserved:** $\eta(z) \equiv D_L / [D_A(1+z)^2] = 1$ exactly
- The effect is **degenerate** with modified $H(z)$ in all distance-only tests
- Degeneracy breaking requires distance-vs-growth consistency (Test 3) or GW comparisons (Test 4)

This is the conservative implementation.

Case (ii): Optical-medium implementation (refractive index)

c_{eff} enters as a spacetime refractive index $n(a) = c_0/c_{\text{eff}}(a)$, where the entropic state of the vacuum acts as a dispersive medium. In this case:

- **CDDR can be violated:** $\eta(z) \neq 1$
- The predicted violation has a **sign change near $z \sim 0.44$** (where T peaks) — a unique EFC fingerprint
- Requires an explicit photon-transport model specifying how $n(a)$ modifies Etherington reciprocity

For $\eta \neq 1$ to follow, one needs: (a) photons not following null geodesics of *any* metric, or (b) effective photon number non-conservation, or (c) explicit breaking of reciprocity conditions.

Current EFC position: EFC's language ("entropy as the medium's hidden variable") points toward a medium-like interpretation, but the formalism has not committed. **This is an open theoretical question that must be resolved before CDDR becomes a smoking gun.**

3.3 CDDR constraint (Case ii only)

Recent CDDR tests using DESI DR2 + Pantheon+ find η consistent with 1 at $\sim 2\%$ (arXiv:2506.17926, arXiv:2505.15768). In Case (ii), this constrains $|\epsilon \cdot \beta \cdot \Delta T| \lesssim 0.02$, giving $\epsilon \lesssim 1.9$. This is weak. Future sub-percent CDDR at multiple redshifts could probe $\epsilon \sim 0.5$.

Important caveat: The $\eta(z) = c_{\text{eff,emit}}/c_{\text{eff,obs}}$ relation requires a specific photon transport model. Until EFC derives this, the prediction is conditional.

4. Degeneracy-Breaking Tests

4.1 Test 1: CDDR (Case ii only)

In the optical-medium implementation, w_0w_a CDM preserves CDDR exactly ($\eta = 1$), while EFC with $\varepsilon \neq 0$ violates it with a sign change at $z \sim 0.44$. This is a unique fingerprint not reproducible by standard dark energy. Requires: (1) derivation that EFC's medium yields $\eta \neq 1$; (2) CDDR at $\sim 1\%$ precision at multiple redshifts.

4.2 Test 2: ISW kernel shape (second-order probe)

If c_{eff} varies, the ISW kernel gains an additional term from $\partial c_{\text{eff}}/\partial t$, modifying the shape of C_ℓ^{Tg} at $\ell < 60$. This is a second-order probe, currently noise-limited at relevant multipoles. Future test with DESI $\times$ Planck tomography.

4.3 Test 3: β -split (distance vs growth) — works in both cases

Why it works: If $\varepsilon \neq 0$, distance-derived β absorbs the c_{eff} effect (since $d_C \propto c_{\text{eff}}/H$), while growth-derived β does not (the growth ODE depends on H and G , not on c directly). A split $\beta_{\text{distance}} \neq \beta_{\text{growth}}$ at $>3\sigma$ indicates $\varepsilon \neq 0$.

This is the cleanest test, working regardless of implementation (metric or optical-medium). BAO/SN estimate the “distance sector” which absorbs c_{eff} changes. Growth ($f\sigma_8$) estimates friction/gravitation, largely insensitive to c_{eff} .

Current status: The four-channel paper shows consistent β across BAO and RSD, constraining $\varepsilon \leq O(1)$. Future DESI DR3 + Euclid can tighten this. Any detected split must be tested against early-time calibration consistency to exclude hidden H_0 priors.

4.4 Test 4: Standard sirens (GW vs EM) — works in both cases

Why this is powerful: If the grid coupling is EM-specific, then $D_L^{\text{GW}} \neq D_L^{\text{EM}}$. Because standard sirens are self-calibrating, this test is insensitive to the cosmic distance ladder and directly probes sector-dependent propagation.

The EFC prediction: If the grid coupling is vacuum/EM-specific: $\delta c_{\text{EM}} \neq \delta c_{\text{GW}} \rightarrow$ mismatch. If universal (conformal factor): $\delta c_{\text{EM}} = \delta c_{\text{GW}} \rightarrow$ no mismatch. Distinguishing these tells you *which* sector the grid couples to.

Status: LIGO/Virgo/KAGRA: ~ 10 sirens, $\sim 10\text{--}20\%$ precision per event. Future: Einstein Telescope + Cosmic Explorer $\rightarrow \sim 1\%$ at $z \sim 0.5 \rightarrow \varepsilon \sim 0.1$ testable.

5. Falsification Criteria

ID	Condition	Meaning	Applies to
F-ceff-1	CDDR violation inconsistent with $T(z)$ shape	c_{eff} not grid-driven	Case (ii)
F-ceff-2	β_{BAO} and β_{RSD} agree to $<1\%$ (better data)	$\varepsilon \approx 0$; no measurable variation	Both
F-ceff-3	CDDR violation monotonic (no sign change)	Not EFC-type new physics	Case (ii)
F-ceff-4	Local lab measurement of c variation	EFC predicts no local effect	Both
F-ceff-5	$D_L^{\text{GW}} = D_L^{\text{EM}}$ to sub-percent (sirens)	No EM-specific modification	Both
F-ceff-6	Case (ii) derived but $\eta = 1$ to $<0.5\%$	Case (ii) ruled out	Case (ii)

6. The Honest Assessment

What this parameterization achieves:

- **Gives a number to test.** $c_{\text{eff}}(a) = c_0[1 + \varepsilon\beta T(a)]$ with $\beta = 0.08$ from data and ε as the one free parameter.
- **Identifies β -split (Test 3) and standard sirens (Test 4)** as implementation-independent degeneracy breakers.
- **Identifies CDDR as conditional smoking gun** (Case ii only).
- **Shows current data consistent but not constraining:** $\varepsilon \lesssim O(1)$.
- **Predicts where future data will be decisive:** β -split from DESI DR3+Euclid; GW sirens from Einstein Telescope.

What it does NOT achieve:

- Does not derive ε from first principles.
- Does not prove variable c . $\varepsilon = 0$ remains viable.
- Does not resolve metric vs optical-medium implementation.
- Does not break $H(z)$ -only degeneracy in Case (i) without β -split or sirens.

Canonical statement:

The present results do not constitute evidence for a varying speed of light. However, within the EFC framework, where late-time dynamics arise from a state-dependent vacuum sector, a weak state dependence of the cosmological propagation limit emerges as a consistency requirement rather than an independent assumption. The parameterization $c_{\text{eff}}(a) = c_0[1 + \varepsilon\beta T(a)]$ introduces one additional parameter (ε) whose observable consequences depend on the implementation: in a metric implementation, CDDR is preserved and degeneracy breaking requires distance-vs-growth consistency or gravitational-wave standard sirens; in an optical-medium implementation, CDDR violation with a specific sign-change near $z \sim 0.44$ becomes a unique signature. Resolving this theoretical ambiguity and establishing the effect observationally are the critical next steps.

7. Next Steps

Theoretical (prerequisite for Test 1):

- Resolve metric vs optical-medium: derive from EFC's coupling structure whether the grid modifies the conformal metric (Case i) or acts as an effective refractive medium (Case ii).

Computational:

- Compute β_{distance} vs β_{growth} split as function of ϵ
- Compute $\eta(z)$ numerically for Case (ii) at $\epsilon = 0.1, 0.5, 1.0$
- Check ISW kernel modification with $c_{\text{eff}}(a)$
- Estimate ϵ threshold for Einstein Telescope detection at 2σ

Observational targets:

- DESI DR3 BAO + RSD — tighten β -split constraint
- Rubin/LSST SN Ia — sub-percent CDDR
- LIGO O5 / Einstein Telescope — standard sirens at $z \sim 0.1\text{--}0.5$

References

- [1] Magnusson, M. (2026). Four-channel consistency test. DOI: 10.6084/m9.figshare.31304980
- [2] Magnusson, M. (2026). Regime consistency: $\text{Ly}\alpha$, BAO, RSD. DOI: 10.6084/m9.figshare.31304995
- [3] Magnusson, M. (2025). Variable effective light speed in entropic transition states. DOI: 10.6084/m9.figshare.30642497
- [4] Magnusson, M. (2025). ISW cross-correlation predictions. DOI: 10.6084/m9.figshare.31301953
- [5] Santos et al. (2025). meVSL constraints via CDDR. arXiv:2505.15768
- [6] Zhang et al. (2025). CDDR with DESI DR2. arXiv:2506.17926
- [7] Alfano & Luongo (2025). CDDR after DESI 2024. arXiv:2501.15233
- [8] Teixeira et al. (2025). Distance duality violation implications. arXiv:2504.10464
- [9] DESI Collaboration (2025). Full-shape modeling DR1. arXiv:2411.12022